1. Among the leaders in the early 1980s in determining how quantum fluctuations would yield inhomogeneities were Stephen Hawking, Alexei Starobinsky, Alan Guth, So-Young Pi, James Bardeen, Paul Steinhardt, Michael Turner, Viatcheslav Mukhanov, and Gennady Chibisov.

2. Even with the discussion in the main text, you may still be puzzled regarding how a tiny amount of mass/energy in an inflaton nugget can yield the huge amount of mass/energy constituting the observable universe. How can you wind up with more mass/energy than you begin with? Well, as explained in the main text, the inflaton field, by virtue of its negative pressure, "mines" energy from gravity. This means that as the energy in the inflaton field increases, the energy in the gravitational field decreases. The special feature of the gravitational field, known since the days of Newton, is that its energy can become arbitrarily negative. Thus, gravity is like a bank that is willing to lend unlimited amounts of money—gravity embodies an essentially limitless supply of energy, which the inflaton field extracts as space expands.

The particular mass and size of the initial nugget of uniform inflaton field depend on the details of the model of inflationary cosmology one studies (most notably, on the precise details of the inflaton field's potential energy bowl). In the text, I've imagined that the initial inflaton field's energy density was about 10^{82} grams per cubic centimeter, so that a volume of $(10^{-26}$ centimeters$)^3 = 10^{-78}$ cubic centimeters would have total mass of about 10 kilograms, i.e., about 20 pounds. These values are typical to a fairly conventional class of inflationary models, but are only meant to give you a rough sense of the numbers involved. To give a flavor of the range of possibilities, let me note that in Andrei Linde's chaotic models of inflation (see note 11 of Chapter 10), our observable universe would have emerged from an initial nugget of even smaller size, 10^{-33} centimeters across (the so-called Planck length), whose energy density was even higher, about 10^{94} grams per cubic centimeter, combining to give a lower total mass of about 10^{-5} grams (the so-called Planck mass). In these realizations of inflation, the initial nugget would have weighed about as much as a grain of dust.

3. See Paul Davies, "Inflation and Time Asymmetry in the Universe," in *Nature*, vol. 301, p. 398; Don Page, "Inflation Does Not Explain Time Asymmetry," in *Nature*, vol. 304, p. 39; and Paul Davies, "Inflation in the Universe and Time Asymmetry," in *Nature*, vol. 312, p. 524.

4. To explain the essential point, it is convenient to split entropy up into a part due to spacetime and gravity, and a remaining part due to everything else, as this captures intuitively the key ideas. However, I should note that it proves elusive to give a mathematically rigorous